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COMPUTATIONAL THINKING

Practice 01

APPLICATION PROGRAMMING INTERFACE

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1 Learning Objectives

This assignment aims to help students apply their knowledge of **Web API** and **FastAPI** by deploying a model from **Hugging Face** as a functional API system. Upon completing this assignment, students should be able to:

- Explore, select, and use an appropriate model from Hugging Face.
- Develop a Web API using FastAPI to provide access to the selected model's functionality.
- Organize source code, write documentation for running the application, and manage the project using GitHub.

2 Assignment Description

Each student is required to select **one model** from **Hugging Face** and develop a **Web API** using **FastAPI** to provide access to the model's functionality.
Students may choose from different types of models, including:

- Text classification
- Sentiment analysis
- Text summarization
- Machine translation
- Question answering
- Text generation
- Image classification
- Image captioning
- Speech recognition

3 API Functional Requirements

Students must develop the system as a **FastAPI application** with at least the following endpoints:

- `GET /`: Returns a brief introduction to the system or a description of the API's functionality.
- `GET /health`: Checks the operational status of the system.
- `POST /predict` or `POST /generate`: Receives input data from the user, invokes the Hugging Face model, and returns the results in JSON format.

In addition, the API must satisfy the following requirements:

- Input data must be appropriate for the selected model type.
- The returned results must be clear, well-structured, and formatted as JSON.
- Basic input validation must be implemented.
- Errors such as missing input data, invalid data formats, or failures during model inference must be handled appropriately.

4 Source Code Requirements

Students must submit a **public GitHub repository** containing the complete source code of their project. The repository must include at least the following components:

- The complete source code of the project.
- A `requirements.txt` file specifying the required dependencies.
- A `README.md` file describing the project and providing instructions for running the application.
- At least one Python file for testing the API using the `requests` library.
- A `.ipynb` notebook file if Google Colab or Jupyter Notebook was used during development.

The `README.md` file must clearly include the following information:

- Student information.
- The name of the selected model and a link to its Hugging Face page.
- A brief description of the system's functionality.
- Instructions for installing the required dependencies.
- Instructions for running the application.
- Instructions for calling the API, including example requests and responses.
- A link to the demonstration video.

5 Demonstration Video Requirements

Each student must prepare **one demonstration video** **(Maximum 5 minutes)** showcasing the developed system.

- Introduce the selected Hugging Face model.
- Briefly explain the source code structure.
- Demonstrate how to run the application.
- Test the API using at least two different inputs.
- Explain the results returned by the system.

The demonstration video may be uploaded to YouTube, Google Drive, or a similar platform. If Google Drive is used, students must ensure that the video is accessible through the shared link. The video link must be included in the `README.md` file. Students are encouraged to embed the video link or thumbnail directly in the README to make it easier to access and review.